MILITARY SPECIFICATION

OXYGEN, USP, WITH TUBE AND FACE MASK

24 GAL. (90 LITERS)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a portable, disposable, emergency oxygen apparatus, suitable for the administration of oxygen.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes. Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6505

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this purchase description to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug and Cosmetic Act and General Regulations for its Enforcement.

(Applications should be addressed to the Food and Drug Administration, U. S. Department of Health, Education, and Welfare, Washington, D. C. 20204.)

3. REQUIREMENTS

3.1 Material. Material shall be as herein specified. Material shall be of the highest quality and suitable for the purpose intended.

3.1.1 Oxygen Oxygen shall comply with the requirements, tests, and standards of the current edition of the Pharmacopeia of the United States (U.S.P.).

3.1.2 Cylinder. Cylinder shall be a seamless container fabricated from carbon steel.

3.1.3 Valve. Valve shall be fabricated of a material compatible with the service intended and shall be brass, nickel or chrome plated, or anodized aluminum.

3.1.4 Gasket. Gasket shall be fabricated of materials compatible with oxygen at pressures, temperatures and concentrations specified herein.

3.1.5 Tube, mask, and reservoir bag. The tube, mask, and reservoir bag shall be safe (nontoxic), polyethylene, or vinyl, plastic material suitable for the purpose intended.

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3.2 Style, design, and dimensions. Disposable, emergency oxygen apparatus shall consist essentially of a filled oxygen cylinder, regulator, plastic face mask, plastic reservoir bag, and plastic tubing. Cylinders must be fabricated, charged, and shipped under terms of an Interstate Commerce Commision special permit. Each manufacturer and shipper must have his own permit.

3.2.1 Cylinder. The cylinder shall be not less than 12.00 inches nor more than 18 inches in overall length and not less than 1-15/16 inches nor more than 4 inches in diameter, and shall have a minimum wall thickness of 0.045 inch and a minimum volume of 96 cubic inches (1570 cubic centimeters). The cylinder shall have a minimum burst strength in accordance with Table I below, be pressure-tested in accordance with Table I without evidence of leakage, distortion, or other defect.

When required by the Interstate Commerce Commission, the cylinder shall be equipped with a fusible plug safety device utilizing fuse metal having a yield temperature of 212° Fanrenheit (F.). The fusible plug safetv device shall be capable of preventing rupture of the normally charged cylinder when it is placed in a fire. The complete cylinder shall show no evidence of leakage when tested as specified in 4.4.3.

Table I Burst Strength

Cylinder type	Pounds per square inch	Minimum burst strength	Cylinder pressure at 70° F (lbs per sq. inch)
I	1200	2500 lbs	975 to 1025
II	2250	3600 lbs	1775 to 1825

3.2.2 Regulator. The regulator shall be so constructed as to be an integral part of the unit and shall permit the flow of gas to be turned off without disconnecting the regulator from the unit. The operation of the regulator control knob in a clockwise direction to the maximum extent will permit oxygen to flow at a fixed rate of 6 liters per minute across the entire range of tank pressure (see 3.5).

3.2.3 Oxygen. The cylinder shall contain 90 to 100 liters of oxygen at standard conditions (0° Centigrade C. and 760 millimeters mm. of mercury). The cylinder pressure shall be in accordance with Table I (see 3.2.1).

A contents indicator may be provided which will visually provide indication of oxygen contents across the range of cylinder pressures.

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3.4 Mask, reservoir bag, and tubing. Mask shall be anatomically designed to fit over the mouth and nose, and shall be supplied with a reservoir bag as well as an elastic band to fit around the head. Tubing shall be a minimum of 5 feet in length; outside diameter shall be a minimum of 5/32 inch.

3.3 Weight. The weight of the cylinder, complete with the regulator and oxygen, shall be 2-1/4 pounds minimum.

3.4 Performance characteristics. The unit shall deliver oxygen at a rate of 6 liters per minute minimum at 1000 pounds per square inch gage through 50 pounds per square inch gage at 70° F., or at 1800 p.s.i.g. through 80 p.s.i.g. at 70° F., for 15 (± 1.3) minutes, at 16 liters per minute.

3.5 Safety (nontoxicity).

3.5.1 Tubing. Tubes shall be safe (nontoxic), when tested for safety as specified in 4.4.2.1.

3.5.2 Mask and reservoir bag. When tested as specified in 4.4.2.2, the sample shall comply with the requirements of the U.S.P. for Systemic Injection Test, Plastic Container.

3.6 Certificates of safety (nontoxicity). A certificate of safety (nontoxicity), listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety (nontoxicity), as required in 3.6 and 4.4.2, and have been found to be safe (nontoxic). The certificate shall reference the method used in determining safety (nontoxicity). For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, Technical Operations Division, 2800 South 20th Street, Philadelphia, Pa 19101, ATTN: Quality Control Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.7 Manufacturer's record. The manufacturer shall place a code or lot number on the label, on cylinder and on the printed carton, which can be verified against the manufacturer's records of all tests made on nontoxic masks, reservoir bags, and tubes. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

* 3.8 Finish. Cylinder shall be painted with green or white enamel, and instructions thereon shall be printed with a suitable contrasting color.

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3.9 Identification and labeling. Cylinder shall be legibly and permanently marked with the manufacturer's name or registered trademark. Labeling shall comply with the regulations of the Federal Food, Drug, and Cosmetic Act.

3.10 Instructions. Each unit shall be supplied with an instruction sheet indicating proper use of equipment. In addition, instructions for proper use of equipment shall be printed on the cylinder.

3.11 Workmanship. Workmanship shall be first class throughout. The equipment shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual of auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, plastic, brass or aluminum, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

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4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for examination purposes shall be one emergency oxygen apparatus.

Table II Sampling for Examination

Defect category	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests.

4.2.2.1 For oxygen Sampling for the oxygen test shall be conducted as indicated in the U.S.P.

4.2.2.2 For other tests Sampling for other tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table III. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The unit of product shall be one emergency oxygen apparatus.

Table III Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Cylinder	Dimensions	3.2.1	-	S-2	1.0
Cylinder	Burst strength	3.2.1	4.4.1	S-1	*
Cylinder	Leakage	3.2.1	4.4.2	100%	*
Tubes	Safety	3.5.1	4.4.3.1	**	*
Mask and reservoir bag	Systemic injection	3.5.2	4.4.3.2	**	*

* Acceptance number shall be zero.

** Sampling for tests shall be on component lots.

4.3 Examination. The emergency oxygen apparatus shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table IV. Inspection, however, shall not be restricted to the classified possible defects listed in Table IV. Any emergency apparatus in the sample containing one or more defects shall be rejected. If the number of defective devices in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

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Table IV Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	One or more components missing.
102	Cylinder not of size specified.
103	Weight not as specified.
104	Instructions for proper use incomplete, or illegible.
105	Performance characteristics not as specified.
106	Finish not as specified.
Minor	
201	Packaging, packing and marking not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Burst test. Burst test shall be performed by use of hydrostatic pressure apparatus with the cylinder enclosed in a heavy steel pipe or jacket.

4.4.2 Leakage. The filled cylinder shall be tested for leakage by pouring Winton solution, or equal, around valve and safety device, and allowing it to remain idle for a period of 30 minutes. There shall be no uniform continuous bubbles upon visual inspection.

4.4.3 Safety tests

4.4.3.1 Tubes. Tubes shall be tested for safety in accordance with the U.S.P. for transfusion and infusion assemblies.

4.4.3.2 Mask and reservoir bag. Mask and reservoir bag shall be tested in accordance with the U.S.P. for Systemic Injection Test, Plastic Container.

4.4.4 Oxygen. Oxygen in the filled cylinder shall be tested in accordance with U.S.P. Monograph. Alternate assay method and procedures are not permitted.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1. Unit of issue. One package containing one oxygen cylinder with regulator, face mask, tubing and reservoir bag as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table V Packaging Quantities

Unit Package	Intermediate package	Exterior container
1 unit	Not required	12 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A .

5.1.3.1 Unit package. The face mask, tubing, and reservoir bag shall be assembled. The reservoir bag shall be neatly folded against the mask and the tubing coiled in such a manner as to prevent kinking. The assembled mask, tubing and reservoir bag shall be inserted into a clear plastic bag of appropriate size. The bag shall be closed in such a manner as to prevent spilling of contents and to facilitate easy opening. The cylinder shall also be inclosed in a clear plastic bag of appropriate size and closed in such a manner as to facilitate easy opening. The bagged components along with instruction data shall be packaged in a corrugated fiberboard box of appropriate size and design. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure of the box shall be adequate to prevent spilling of contents under normal handling. Commercial colors are acceptable on unit boxes.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

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5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3, PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. In addition to any commercial markings, each unit package shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and MIL-STD-129.

5.3.2 Exterior container Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The disposable oxygen cylinder is intended for the administration of oxygen under emergency conditions.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6505-965-2439	OXYGEN, USP, Disposable, with Tube and Face Mask, 24 gal (90 liters).

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes requirements of Purchase Description No. 3 dated 10 February 1971.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1279

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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